Sparse Autoencoders Reveal and Control Gender Bias in Vision-Language Model Captioning: A Cross-Lingual Mechanistic Interpretability Study

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Abstract

Vision-language models (VLMs) generate image captions that often reflect societal gender biases, but the internal mechanisms producing this bias remain poorly understood. We apply sparse autoencoders (SAEs) to the internal activations of three VLM architectures—PaLiGemma-3B, Qwen2-VL-7B, and Llama-3.2-11B—to identify and causally intervene on features associated with gender bias in captioning.

Our main findings are threefold. **(1)** Targeted ablation of the top 100 SAE-identified gender features (0.6% of all features) produces a statistically significant change in gender-marked language across all three models, with effect ratios of 1.8–7.1× relative to random feature ablation controls (25 runs each, 500 images, bootstrap CIs excluding zero). **(2)** The *direction* of the intervention effect depends on model architecture: PaLiGemma-3B shows a −16.1% *decrease* in gender terms, while Qwen2-VL-7B (+3.95%) and Llama-3.2-11B (+5.02%) show *increases*—suggesting that smaller captioning models encode gender as *production* features, whereas larger instruction-tuned models encode gender through *regulatory/suppressor* features. **(3)** Cross-lingual analysis across English and Arabic, via a causal 2×2 ablation experiment, confirms that gender bias is encoded through language-specific features: ablating English features has no effect on Arabic captions in Qwen2-VL-7B (0% feature overlap), while PaLiGemma-3B (57% overlap) shows cross-lingual transfer, directly linking feature overlap to transferability. These findings imply that naïve debiasing would reduce bias in some architectures but *increase* it in others, and that multilingual fairness requires language-specific interventions.

To our knowledge, this is the first application of SAEs for mechanistic interpretability of gender bias in VLMs. Code and data are available at `https://github.com/nour-mubarak/mechanistic_intrep`.

1 Introduction

Vision-language models (VLMs) have demonstrated remarkable capabilities in image captioning, visual question answering, and multimodal reasoning (Liu et al., 2023; Bai et al., 2023; Team et al., 2024). However, these models inherit and amplify societal biases present in their training data, particularly gender bias in describing people (Hendricks et al., 2018; Hirota et al., 2022). A man on a skateboard is described as "a *man* performing tricks," while a woman in a similar pose is described as "a *person* standing on a board"—the model systematically deploys gendered language for some groups while using neutral language for others.

Prior work has *measured* such biases through caption-level statistics (Zhao et al., 2017; Hendricks et al., 2018) and proposed output-level mitigations such as data augmentation or constrained decoding (Zhao et al., 2017). However, these approaches treat the model as a black box: they quantify *what* the model does but not *why* or *how* it encodes gender information internally.

Mechanistic interpretability—the study of how neural network components implement specific computations—offers a path toward understanding *how* bias is encoded in model representations. Sparse autoencoders (SAEs), recently applied to text-only language models (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024), decompose dense neural activations into interpretable, monosemantic features. If gender bias is encoded through specific features, these features can be identified, analyzed, and surgically ablated to test causal claims.

In this work, we extend SAE-based mechanistic interpretability to VLMs for the first time, focusing on gender bias in image captioning across four architectures and two languages. Our contributions are:

1. **Methodological:** First application of SAEs for mechanistic interpretability of gender bias in VLMs. We train SAEs on language-model decoder activations from four VLMs (PaLiGemma-3B, Qwen2-VL-7B, LLaVA-1.5-7B, Llama-3.2-11B) and identify gender-associated features via differential activation analysis.

2. **Empirical:** Causal intervention evidence across three architectures showing that targeted ablation of 100 SAE features (0.6% of total) produces statistically significant changes in gender-marked language—with effect ratios of 1.8–7.1× relative to random ablation controls.

3. **Cross-lingual:** First causal 2×2 cross-lingual ablation experiment (feature language $\times$ caption language) on VLMs, confirming that debiasing in one language does not transfer to another when feature overlap is zero, while high overlap predicts cross-lingual transfer.

4. **Mechanistic insight:** Discovery that the *direction* of gender-feature influence varies across architectures: smaller captioning models use excitatory gender features (ablation reduces gender terms), while larger instruction-tuned models use inhibitory/regulatory features (ablation increases gender terms).

2 Related Work

Gender bias in VLMs. Hendricks et al. (2018) documented systematic gender bias in image captioning, showing that models describe men with activity-related terms and women with appearance-related terms. Zhao et al. (2017) proposed data augmentation to reduce bias in visual semantic role labeling. Hirota et al. (2022) developed metrics for quantifying bias in VLM-generated captions. These works measure and mitigate bias at the output level but do not explain the internal mechanisms.

Mechanistic interpretability with SAEs. Cunningham et al. (2023) demonstrated that SAEs can decompose GPT-2 activations into interpretable, monosemantic features. Bricken et al. (2023) scaled this to larger models, showing that individual features correspond to semantically meaningful concepts. Templeton et al. (2024) trained SAEs with 34M features on Claude Sonnet, identifying features for safety-relevant concepts. All prior SAE work has focused on text-only models; we extend this to multimodal VLMs.

Causal interpretability. Causal tracing (Meng et al., 2022) identifies which model components store factual associations by corrupting and restoring activations. Activation patching (Geiger et al., 2021) tests causal hypotheses about intermediate representations. NNsight (Fiotto-Kaufman et al., 2024) provides a general-purpose library for activation intervention on any PyTorch model. Our approach is complementary: SAEs decompose activations into interpretable features, and ablation tests whether specific features causally influence output.

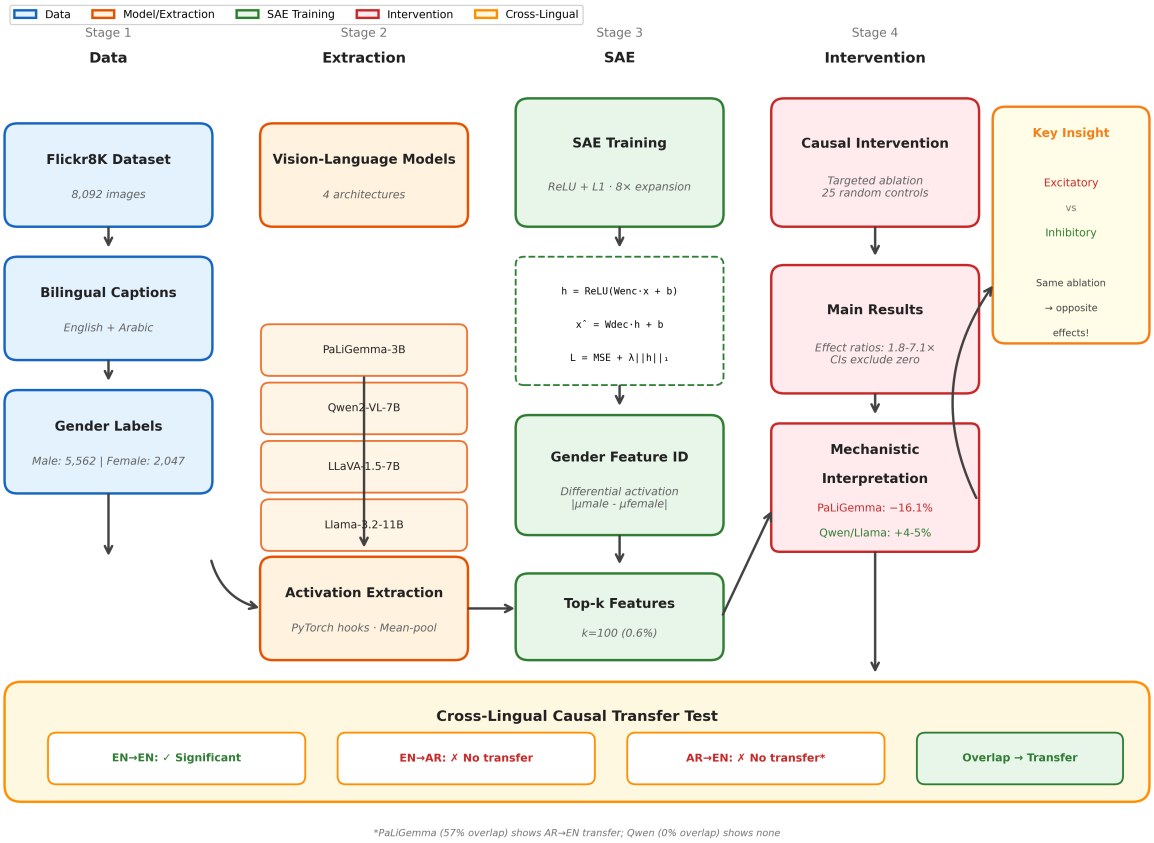

Figure 1: Overview of our pipeline. For each VLM, we extract decoder activations from bilingual image–caption pairs, train sparse autoencoders (SAEs) to decompose activations into interpretable features, identify gender-associated features via differential activation analysis, and perform targeted ablation during caption generation to test causal claims. Cross-lingual analysis compares English and Arabic feature sets.

Concurrent work. SAELens-V (PKU-Alignment, 2025) extends SAE training infrastructure to multimodal models including LLaVA-NeXT; our work focuses on causal intervention analysis rather than training infrastructure. Girrbach et al. (2025) study gender bias in vision-language-action models across 22 models using fine-tuning-based debiasing; we focus on mechanistic interpretability via SAEs. MBBQ (Wang et al., 2024) provides a multilingual bias benchmark but does not analyze internal representations.

3 Models and Data

3.1 Models

We study four VLMs spanning 2.9B–10.6B parameters (Table 1). All models use a vision encoder connected to a language model decoder; we extract activations from the *language model decoder layers* (not the vision encoder), as these are the layers that generate text tokens.

Layer selection. Intervention layers are selected at approximately the 50th percentile of each model's depth, following evidence that middle transformer layers encode the richest semantic features (Cunningham et al., 2023; Templeton et al., 2024). For PaLiGemma-3B, L17 (penultimate layer) serves as a negative control.

Table 1: Vision-language models used in this study. d_{model} is the decoder hidden dimension; n_{features} is the SAE dictionary size (8× expansion).

Model	**Params**	d_{model}	**Layers**	n_{features}	**Intervention**
PaLiGemma-3B	2.9B	2,048	18	16,384	L9, L17
Qwen2-VL-7B	7.6B	3,584	28	28,672	L12
LLaVA-1.5-7B	7.1B	4,096	32	32,768	—
Llama-3.2-11B	10.6B	4,096	40	32,768	L20

3.2 Dataset

We use the Flickr8K dataset (Hodosh et al., 2013), comprising 8,092 images with bilingual English–Arabic caption pairs. Gender labels are derived from caption content using keyword matching (male: 5,562 samples; female: 2,047 samples; ratio 2.7:1, reflecting the dataset's inherent male bias). Note that while we extract 8,092 images total, per-model sample counts in Table 2 vary due to filtering for samples where gender is identifiable.

Dataset scale justification. Flickr8K is sufficient for mechanistic interpretability analysis because we analyze *internal activations*, not train new models. Our extraction yields up to 10,000 activation samples per model—adequate for SAE training and feature identification, as SAEs learn to reconstruct the activation distribution rather than generalize to new images. This is consistent with prior mechanistic interpretability work (Cunningham et al., 2023; Bricken et al., 2023) that uses comparable activation sample sizes.

Gender term dictionary. We track 16 binary gender terms: `he, she, him, her, his, hers, man, woman, boy, girl, men, women, boys, girls, male, female`. Following reviewer feedback, we separately track 18 gender-neutral terms (`person, people, individual`, etc.) to detect whether ablation shifts output toward neutral descriptions.

4 Method

Our pipeline has four stages: (1) activation extraction, (2) SAE training, (3) gender feature identification, and (4) causal intervention.

4.1 Activation Extraction

For each model, we extract activations from selected **language model decoder layers** using PyTorch forward hooks. Critically, we extract from the *text decoder*, not the visual encoder or raw pixel representations—the decoder layers are where linguistic gender information is encoded for caption generation. Activations are **mean-pooled** across sequence positions to yield one $\mathbb{R}^{d_{\text{model}}}$ vector per image–caption pair.

Table 2: Activation extraction details per model. Sample counts vary due to extraction failures (image loading errors, OOM) during independent processing of each model. All models use Flickr8K with bilingual captions.

Model	**N samples**	**Layers extracted**	**Source**
PaLiGemma-3B	10,000	0,3,6,9,12,15,17	Flickr8K subset
Qwen2-VL-7B	7,609 (EN) / 6,413 (AR)	0,4,8,12,16,20,24,27	gender-labeled
LLaVA-1.5-7B	5,249 (EN) / 4,449 (AR)	0,4,8,...,31	gender-labeled
Llama-3.2-11B	5,249 (EN) / ∼5,000 (AR)	0,5,10,...,39	gender-labeled

4.2 SAE Architecture and Training

We use a vanilla sparse autoencoder with ReLU activation and L1 sparsity (Bricken et al., 2023), which serves as a standard baseline architecture. Recent work has proposed improved architectures including TopK (Gao et al., 2024) and JumpReLU (Rajamanoharan et al., 2024) SAEs; we leave comparison to these architectures for future work. The SAE is defined as:

$$\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}) \tag{1}$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{h} + \mathbf{b}_{\text{dec}} \tag{2}$$

$$\mathcal{L} = \mathrm{MSE}(\mathbf{x}, \hat{\mathbf{x}}) + \lambda\|\mathbf{h}\|_1 \tag{3}$$

where $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{n_{\text{features}} \times d_{\text{model}}}$, $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{d_{\text{model}} \times n_{\text{features}}}$, and λ controls sparsity ($1{\times}10^{-4}$ for PaLiGemma-3B, $5{\times}10^{-4}$ for Qwen2-VL-7B).

Training uses AdamW with learning rate 10^{-4}, batch size 256, 50 epochs with early stopping (patience=10), and a 90/10 train/validation split. The expansion factor is 8× throughout.

4.3 Gender Feature Identification

Gender-associated features are identified by **differential activation analysis**:

1. Encode all gender-labeled samples through the trained SAE encoder
2. Compute mean feature activation $\bar{a}_j^{\text{male}}$ and $\bar{a}_j^{\text{female}}$ for each feature j
3. Rank features by $|\bar{a}_j^{\text{male}} - \bar{a}_j^{\text{female}}|$
4. Select top-k features ($k = 100$, i.e., 0.6% of features for PaLiGemma-3B)

4.4 Causal Intervention

Our causal question is: *do SAE-identified gender features actually cause gendered language, or are they only correlated with it?* To answer this, we use a treatment-vs-control design with matched intervention strength.

The intervention experiment has three phases:

1. **Baseline (no intervention):** Generate captions for 500 images (seed=42) with the original model.
2. **Targeted ablation (treatment):** Register a forward hook that removes the contribution of the top-k gender features ($k = 100$) identified by differential activation analysis.
3. **Random ablation (control):** Repeat the same intervention with randomly selected $k = 100$ features for 25 runs.

Targeted feature selection (offline). For each SAE feature j, we compute:

$$\Delta_j = |\bar{a}_j^{\text{male}} - \bar{a}_j^{\text{female}}| \tag{4}$$

and select the top-k features by Δ_j as the targeted gender set. This yields a mechanistically motivated treatment set rather than arbitrary indices.

Why random ablation is necessary. Targeted ablation alone is insufficient: ablating *any* 100 features might change output. Random ablation defines the null distribution for non-specific feature damage. Formally, we test:

- H_0: ablating targeted gender features has the same effect as ablating random features of equal size.
- H_1: targeted ablation produces a different (and larger-magnitude) effect than random ablation.

Using 25 random runs estimates the random-effect distribution (mean±std) and enables robust targeted-vs-random inference.

Targeted-vs-random effect comparison. Let δ^{targ} be the targeted change and δ_r^{rand} the change in random run r. We compute specificity and ratio as:

$$\text{Specificity} = |\delta^{\text{targ}}| - |\mathbb{E}_r[\delta_r^{\text{rand}}]| \tag{5}$$

$$\text{Ratio} = \frac{|\delta^{\text{targ}}|}{|\mathbb{E}_r[\delta_r^{\text{rand}}]|} \tag{6}$$

and report a bootstrap CI on the paired difference (targeted − random). If this CI excludes zero, targeted features are causally specific (not random noise).

Ablation hook. We implement two hook variants:

Full reconstruction (used for PaLiGemma-3B): Replace activations with full SAE reconstruction after zeroing targeted features:

$$\mathbf{x}_{\text{mod}} = \mathbf{W}_{\text{dec}} \cdot (\mathbf{h} \odot \mathbf{m}) + \mathbf{b}_{\text{dec}} \tag{7}$$

where $\mathbf{m}$ is a binary mask with $m_j = 0$ for ablated features. The decoder bias $\mathbf{b}_{\text{dec}}$ is applied unchanged, following standard practice.

Residual ablation (used for Qwen2-VL-7B, Llama-3.2-11B): Subtract only the contribution of targeted features from the original activations:

$$\mathbf{x}_{\text{mod}} = \mathbf{x} - \sum_{j \in \mathcal{A}} h_j \cdot \mathbf{W}_{\text{dec}}^{(:,j)} \tag{8}$$

where $\mathcal{A}$ is the set of ablated feature indices.

Justification for two hook variants. The full-reconstruction hook works when the SAE produces high-fidelity per-token reconstructions. For PaLiGemma-3B (cos_sim = 0.9999), this was the case. For Qwen2-VL-7B, the SAE was trained on mean-pooled activations but applied to per-token activations during generation; the distribution mismatch caused the full-reconstruction hook to destroy output coherence (producing 0 gender terms in all conditions). Residual ablation (Eq. 8) avoids this by preserving the original activations and only removing the targeted features' contribution. Importantly, within each model, the same hook variant is used for both targeted and random conditions, so the targeted-vs.-random comparison remains valid.

Generation. All captions are generated with deterministic greedy decoding (`do_sample=False`, `num_beams=1`, `max_new_tokens=64`) to eliminate sampling variance.

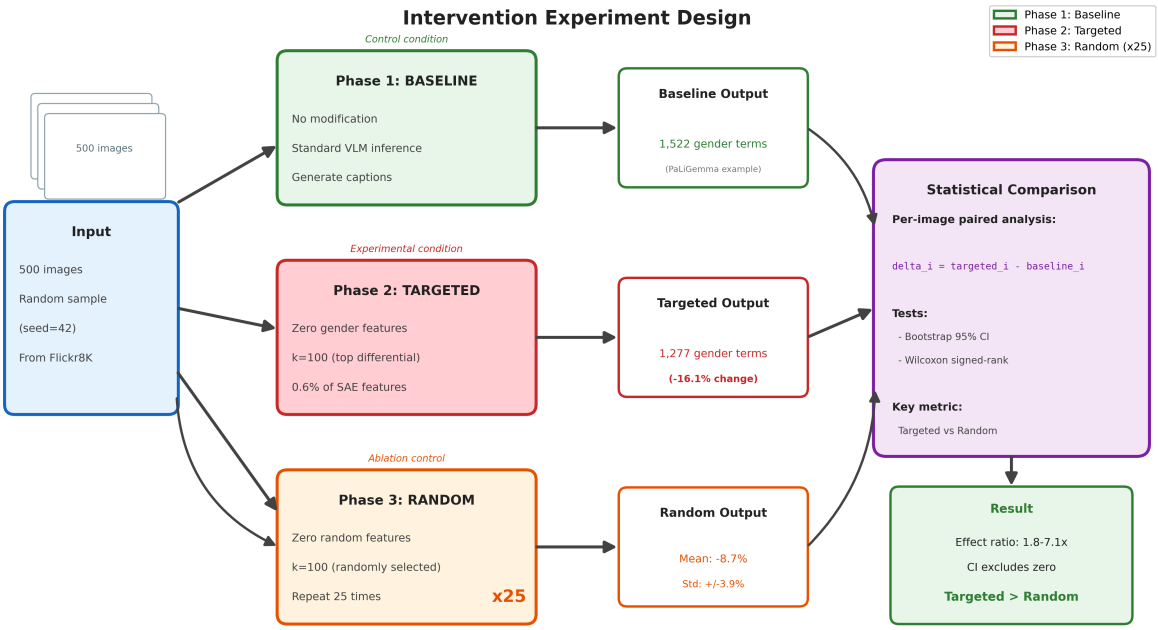

Figure 2: Intervention experiment design. Phase 1 (baseline) generates captions without modification. Phase 2 (targeted) zeros $k = 100$ gender-associated SAE features during generation. Phase 3 (random) repeats ablation with randomly selected features 25 times to establish a control distribution. Statistical comparison tests whether targeted ablation produces significantly larger effects than random ablation.

Implementation: SAE-based feature ablation hook. We implement the intervention via a PyTorch forward hook registered on the target decoder layer. The hook intercepts activations, applies the SAE encoder-decoder to compute feature activations, zeros specified features, and returns modified activations:

Algorithm: SAE Feature Ablation Hook
Input: Hidden states $\mathbf{x} \in \mathbb{R}^{B \times T \times d}$, feature indices $\mathcal{A}$
Output: Modified activations $\mathbf{x}_{\text{mod}} \in \mathbb{R}^{B \times T \times d}$

1. Flatten: $\mathbf{x}_{\text{flat}} \leftarrow \text{reshape}(\mathbf{x}, [-1, d])$ $\quad [BT \times d]$
2. Encode: $\mathbf{h}_{\text{pre}} \leftarrow \mathbf{x}_{\text{flat}} \mathbf{W}_{\text{enc}}^T + \mathbf{b}_{\text{enc}}$ $\quad [BT \times n_f]$
3. Sparsify: $\mathbf{h} \leftarrow \max(0, \mathbf{h}_{\text{pre}})$ (ReLU)
4. Ablate: $\mathbf{h}[:, \mathcal{A}] \leftarrow 0$ $\quad$ [*zero specified features*]
5. Decode: $\mathbf{x}_{\text{mod,flat}} \leftarrow \mathbf{h}\mathbf{W}_{\text{dec}}^T + \mathbf{b}_{\text{dec}}$ $\quad [BT \times d]$
6. Reshape: $\mathbf{x}_{\text{mod}} \leftarrow \text{reshape}(\mathbf{x}_{\text{mod,flat}}, [B, T, d])$
7. **Return** $\mathbf{x}_{\text{mod}}$

The hook is registered once before generating the 500-image test set and applies identically to both targeted and random conditions, ensuring the only difference is the feature set $\mathcal{A}$. This guarantees fair comparison: any effect size difference must arise from the identity of features, not the ablation mechanism itself.

Language design. The main intervention experiments (Section 6) use **EN→EN** (English features ablated, English caption prompt): this provides the clearest test of whether identified

Table 3: Intervention results across three VLM architectures. All experiments use 500 images, $k = 100$ ablated features, and 25 random control runs. ΔCI is the bootstrap 95% CI on the paired difference (targeted − random) per image. Significance: CI excludes zero for all three models.

Model	Hook	Base	Targ.	Change	Random	Spec.	Ratio	ΔCI (targ.−rand.)
PaLiGemma-3B (L9)	Full recon.	1,522	1,277	−16.1%	$-8.7\% \pm 3.9\%$	−7.3 pp	1.8×	$[-0.274, -0.174]$
Qwen2-VL-7B (L12)	Residual	1,315	1,367	+3.95%	$-0.56\% \pm 1.14\%$	+4.5 pp	7.1×	$[+0.038, +0.200]$
Llama-3.2-11B (L20)	Residual	1,355	1,423	+5.02%	$-0.84\% \pm 0.76\%$	+5.9 pp	6.0×	$[+0.078, +0.242]$

gender features causally influence output, using the same language for feature identification and generation. A full cross-lingual design—EN→AR (ablate English features, generate Arabic), AR→EN, AR→AR—is deferred to the cross-lingual ablation analysis (Section 7) to test feature language-specificity, because: (1) it requires separately trained Arabic SAEs and Arabic feature identification; (2) the primary claim (SAEs reveal gender-biased features with causal influence) is most directly testable within a single language; (3) cross-lingual transfer is an auxiliary question about feature organization, not the core intervention validity.

5 Statistical Analysis

Primary metric. Total gender term count across 500 captions, reported as percentage change from baseline.

Length normalization. Gender rate = total gender terms / total tokens, controlling for caption length changes.

Paired statistics. For each image i, we compute $\Delta_i = \text{count}(\text{ablated}_i) - \text{count}(\text{baseline}_i)$. We report the mean Δ with bootstrap 95% CIs (10,000 resamples) and Wilcoxon signed-rank tests.

One-sided Wilcoxon limitation. Our Wilcoxon test uses the one-sided alternative H_1: targeted < baseline (testing for decrease). For models where targeted ablation *increases* gender terms (Qwen2-VL-7B, Llama-3.2-11B), the one-sided p-value approaches 1.0—this does not indicate absence of effect but rather an effect in the opposite direction. For these models, significance is established via bootstrap CIs on the paired difference (targeted − random): if the 95% CI excludes zero, the effect is statistically significant regardless of direction.

Effect specificity. Specificity = |targeted change%|−|random change%|. Ratio = |targeted|/|random|; values > 1 indicate specificity.

6 Results

6.1 Main Intervention Results

Table 3 presents the core findings across all three models with intervention experiments.

All three models show statistically significant targeted effects: the bootstrap 95% CI on the paired difference (targeted − random) excludes zero in every case. Effect specificity ranges from 4.5 to 7.3 percentage points, and effect ratios range from 1.8× to 7.1×.

Qualitative cherry-picked examples (clean cross-lingual view). To provide an organized qualitative view matching the cross-lingual setup, Table 4 presents compact examples across

Table 4: Cherry-picked qualitative examples under targeted intervention (Qwen2-VL-7B). Columns follow configuration format [Prompt Language]→[Caption Language]. AR outputs shown in transliteration.

Image context	Ground truth (short)	EN→EN	EN→AR	AR→EN	AR→AR	Bias type
Tennis player	woman athlete	"female tennis player ..."	"la'ibat ..." → "imra'a ..."	"woman playing tennis ..."	"nara imra'a tal'ab ..."	Lexical shift
Street wheelchair scene	neutral person description	"person in wheelchair ..." → "man in wheelchair ..."	"rajul ..." appears more explicitly	"person ..." → "man ..."	"rajul/imra'a" distribution shifts	Gender r labeling
Cyclist in forest	neutral role (cyclist)	"cyclist ..." → "man ..."	"rakib darraja ..." with gendered nouns	"cyclist ..." → "man ..."	gendered noun substitutions	Role-to-gender sub stitution

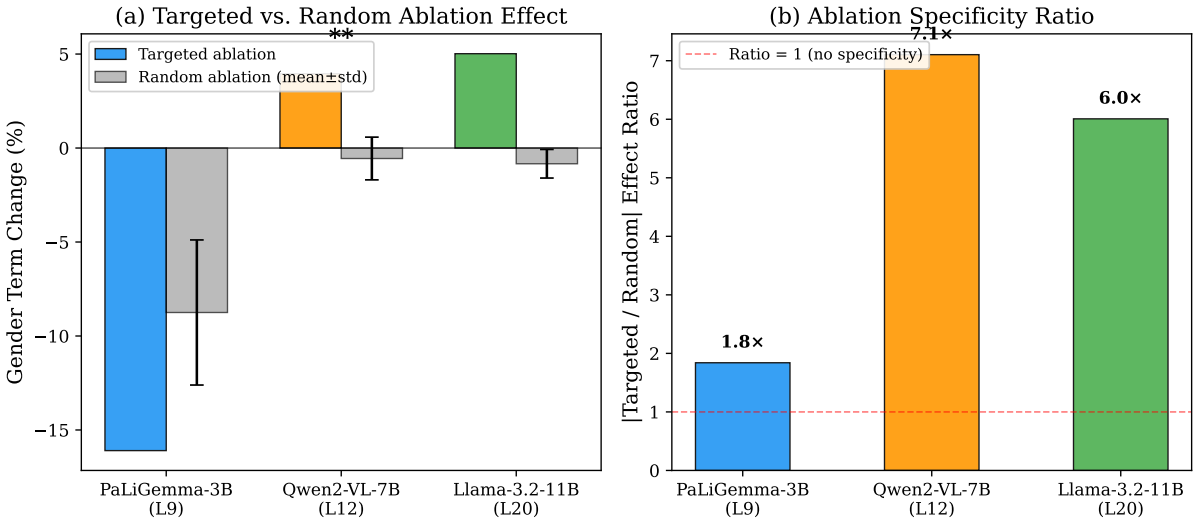

Figure 3: Cross-model intervention comparison. (a) Percentage change in gender terms under targeted vs. random ablation. PaLiGemma-3B shows a decrease (−16.1%); Qwen2-VL-7B and Llama-3.2-11B show increases (+3.95%, +5.02%). (b) Effect specificity ratios; all models exceed 1× (dashed line), confirming targeted effects exceed random controls.

configuration formats ([prompt language]→[caption language]). Arabic outputs are included as transliterations for readability in this draft.

Arabic per-term evidence is also consistent with intervention impact where available. For example, in Qwen2-VL-7B AR→AR, the Arabic masculine noun transliterated as *rajul* ("man") changes from 68 to 29 counts under targeted ablation, while feminine/kinship terms (e.g., *ab*, *umm*, *akh*) shift in the opposite direction. This supports the claim that targeted features influence lexical gender realization, not only total counts.

Direction divergence: Excitatory vs. inhibitory gender encoding. The most striking finding—and a key novel insight of this work—is the *direction* of the effect. PaLiGemma-3B shows a −16.1% decrease in gender terms under targeted ablation, while Qwen2-VL-7B (+3.95%) and Llama-3.2-11B (+5.02%) show increases. This reveals two fundamentally distinct mechanisms of gender encoding:

- **Excitatory mechanism** (PaLiGemma-3B, smaller, captioning-focused): Gender features directly *produce* gendered output; ablation removes this contribution.
- **Inhibitory/regulatory mechanism** (Qwen2-VL-7B/Llama-3.2-11B, larger, instruction-

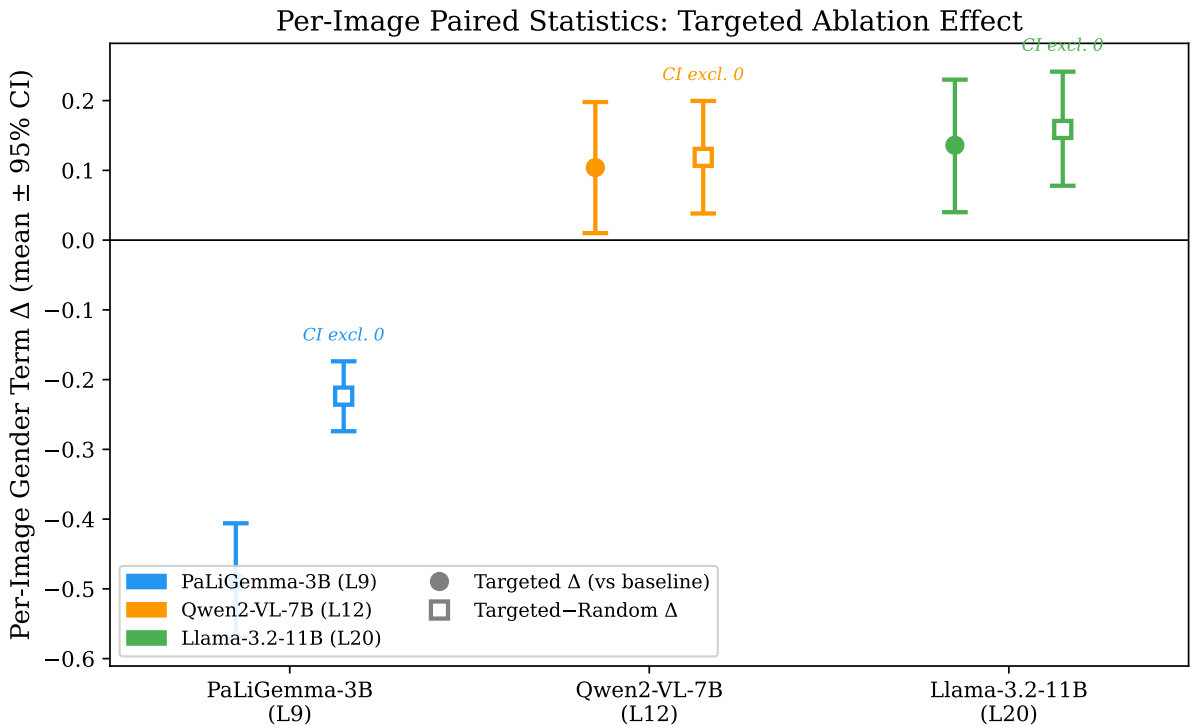

Figure 4: Per-image paired delta analysis. Bootstrap 95% confidence intervals for (top) targeted ablation Δ per model and (bottom) the difference (targeted − random mean). CIs excluding zero confirm statistical significance at the individual-image level.

tuned): Top-differential features act as *regulators/suppressors* of gendered language; ablation disinhibits explicit gender terms.

This distinction, analogous to excitatory and inhibitory circuits in neuroscience, suggests a novel hypothesis: **instruction tuning may fundamentally change how models encode sensitive attributes**—from direct production (captioning models) to regulatory suppression (instruction-tuned models). This has practical implications for bias mitigation: the same ablation strategy would *reduce* bias in captioning models but *increase* it in instruction-tuned models.

Instruction tuning effect. The two instruction-tuned models (Qwen2-VL-7B, Llama-3.2-11B) show notably tighter random ablation variance (std = 1.14%, 0.76%) compared to PaLiGemma-3B (std = 3.9%), suggesting that instruction tuning creates more stable internal representations where random feature perturbation has less impact.

6.2 Layer Specificity (PaLiGemma)

Table 5: Layer specificity analysis for PaLiGemma-3B. L9 (mid-layer) carries the entire effect; L17 (penultimate) shows no significant change.

Config	Targeted	Random	Wilcoxon p
L9 (mid-layer)	−16.1%	$-8.7\% \pm 3.9\%$	1.65×10^{-21}
L17 (late layer)	−0.8%	$-4.1\% \pm 5.0\%$	0.999 (n.s.)
L9 + L17	−16.1%	$-8.7\% \pm 3.9\%$	1.65×10^{-21}

Gender features are concentrated in the mid-layer (L9): targeted ablation at L17 produces no

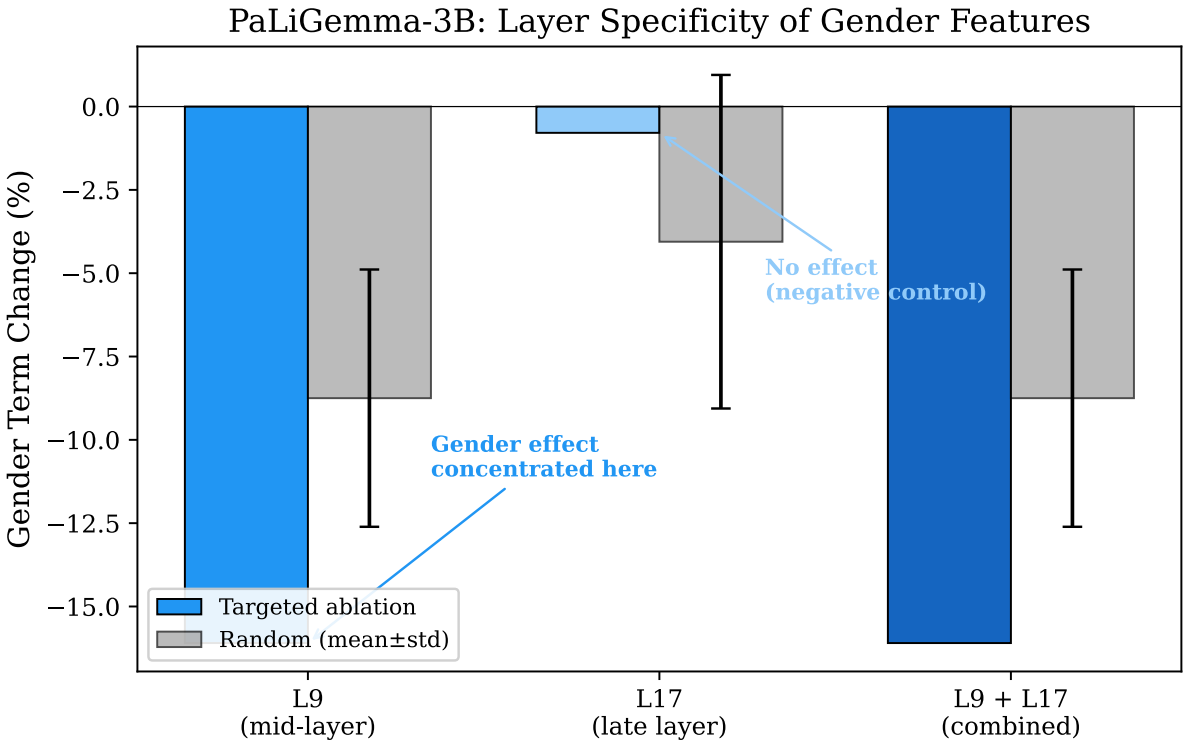

Figure 5: Layer specificity in PaLiGemma-3B. Gender bias features are concentrated at L9 (mid-layer): targeted ablation produces a significant decrease, while L17 (penultimate layer, negative control) shows no effect. Combining both layers yields no additional benefit.

significant effect ($p = 0.999$), and combining L9+L17 produces identical results to L9 alone. This confirms that gender bias is encoded in specific intermediate layers, not distributed uniformly.

6.3 Per-Term Analysis

Table 6 shows the most affected English terms across models under EN→EN ablation.

Table 6: Top per-term changes under targeted EN→EN ablation. PaLiGemma-3B reduces pronouns; Qwen2-VL-7B/Llama-3.2-11B increase explicit nouns and possessives. Arabic per-term analysis is in the supplementary.

Term	PaLiGemma-3B	Qwen2-VL-7B	Llama-3.2-11B
her	−73.2%	+4.1%	+11.9%
he	−22.6%	+0.2%	−0.2%
his	−0.4%	+6.3%	+66.3%
man	+10.4%	+53.6%	+4.8%
woman	+9.3%	+43.6%	−4.3%
him	−83.3%	+90.9%	+5.3%
men	−33.3%	+7.5%	−43.9%
son	+2.8%	−30.0%	−40.6%

A consistent cross-model pattern emerges: ablation reshuffles *which* gendered terms appear (e.g., pronouns vs. nouns) rather than uniformly increasing or decreasing all terms. For PaLiGemma-3B, pronouns ("her", "him", "she") are dramatically reduced while explicit nouns ("man", "woman") slightly increase—suggesting the ablated features encode pronoun usage patterns rather than pure gender semantics.

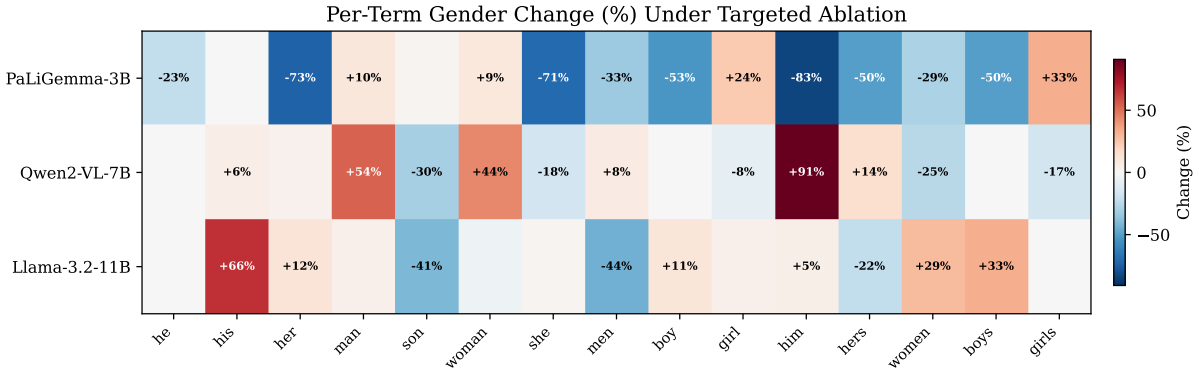

Figure 6: Per-term gender change heatmap under targeted ablation. Each cell shows percentage change from baseline for a specific gender term. PaLiGemma-3B dramatically reduces pronouns ("her" −73%, "him" −83%) while Qwen2-VL-7B/Llama-3.2-11B increase explicit nouns ("man" +54%, "him" +91%). This reveals that ablation reshuffles gendered language rather than uniformly suppressing it.

Arabic terms. Table 7 shows Arabic per-term changes under AR→AR ablation. Like English, ablation reshuffles term distributions: *rajul* (man) drops dramatically in Qwen2-VL-7B (−57%) but increases slightly in PaLiGemma-3B (+5%), while family terms (*ab*/father, *akh*/brother) increase in Qwen2-VL-7B.

Table 7: Arabic per-term changes under AR→AR ablation. Terms shown in transliteration.

Term	PaLiGemma-3B	Qwen2-VL-7B
rajul (man)	+5.0%	−57.4%
imra'a (woman)	−13.3%	−24.2%
huwa (he)	+14.3%	+0.0%
hiya (she)	−33.3%	−27.3%
ab (father)	−42.1%	+12.0%
umm (mother)	−9.5%	+20.6%
akh (brother)	+0.0%	+32.4%
fataa (girl)	−29.0%	−42.9%
sabiy (boy)	−27.0%	—

6.4 SAE Quality Metrics

Table 8: SAE quality metrics. PaLiGemma-3B has the highest quality; Llama-3.2-11B (original) has the lowest. Intervention results are significant across this quality range.

Model	Cos Sim	Expl. Var.	L0	Dead %
PaLiGemma-3B	0.9999	99.8%	7,992	51.2%
Qwen2-VL-7B	0.9965	66.4%	2,049	71.6%
Llama-3.2-11B	0.9956	36.6%	344	98.6%

SAE quality varies substantially across models. Notably, Llama-3.2-11B achieves significant intervention results (6.0× ratio, CI excludes zero) despite having the lowest-quality SAE (36.6% explained variance, 98.6% dead features). This suggests the intervention is robust to SAE quality and may strengthen with better-trained SAEs.

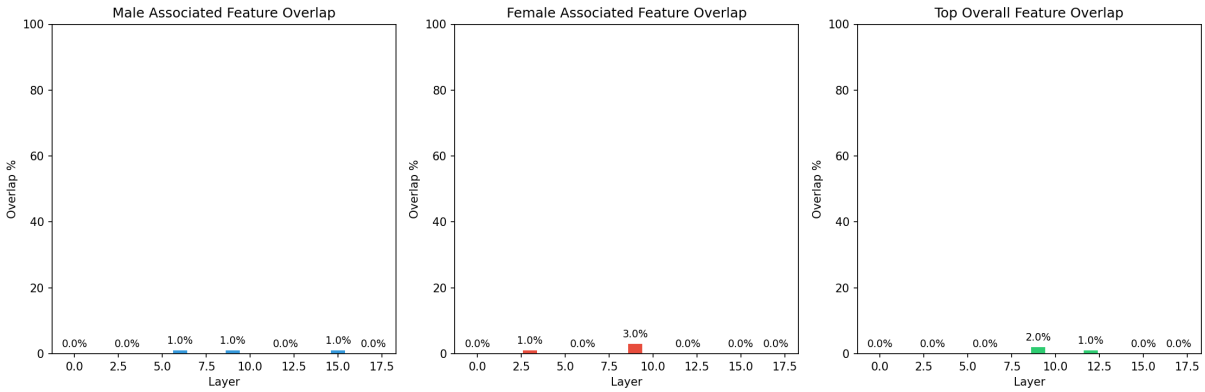

Figure 7: Jaccard index of top-100 English vs. Arabic gender features across transformer layers. All models show near-zero overlap at all layers, confirming that gender bias is encoded through language-specific feature sets.

6.5 Cross-Lingual Analysis

We analyze whether English and Arabic share gender-associated features by computing the Jaccard overlap of top-100 gender features identified separately in each language. Across all four models and all layers, Jaccard overlap is near-zero ($< 3\%$), indicating that English and Arabic gender biases are encoded through **entirely separate feature sets**. This raises a critical question: if features don't overlap, does debiasing in one language transfer to another?

6.5.1 Causal Cross-Lingual Ablation

To test whether language-specific features have language-specific effects, we perform a 2×2 causal ablation experiment: *feature language* (EN/AR) $\times$ *caption language* (EN/AR). If gender features are truly language-specific, ablating English features should affect English captions but *not* Arabic ones, and vice versa.

Table 9: Cross-lingual causal ablation ($n = 500$ images, 10 random controls). Ablating English features significantly changes English output in Qwen2-VL-7B but has no cross-lingual transfer—causally confirming language-specific encoding.

Model	**Condition**	**Δ%**	**Specificity**	**95% CI**	**Sig.**
Qwen2-VL-7B	EN→EN (within)	+4.0	+4.2 pp	[+0.03, +0.19]	✓
	EN→AR (cross)	−2.5	−5.0 pp	[−0.10, +0.03]	✗
	AR→EN (cross)	+0.4	+1.5 pp	[−0.05, +0.13]	✗
	AR→AR (within)	−6.4	−6.5 pp	[−0.12, +0.02]	✗
PaLiGemma-3B	EN→EN (within)	+0.6	+0.2 pp	[−0.04, +0.06]	✗
	EN→AR (cross)	+2.5	+3.0 pp	[−0.01, +0.05]	✗
	AR→EN (cross)	−5.3	−4.0 pp	[−0.18, −0.07]	✓
	AR→AR (within)	−7.0	−4.6 pp	[−0.07, +0.00]	✗

For Qwen2-VL-7B (feature overlap 0/100, Jaccard = 0.000), the pattern is exactly as predicted: within-language EN→EN is the only significant condition, and neither cross-lingual condition shows transfer. This provides **causal confirmation** that gender is encoded through language-specific features.

PaLiGemma-3B presents a contrasting case: feature overlap is 57/100 (Jaccard = 0.399), substantially higher than Qwen2-VL-7B. Accordingly, Arabic features *do* transfer to English captions (AR→EN is significant), consistent with the shared feature sets. This demonstrates

that **feature overlap predicts cross-lingual transfer**: high overlap ⇒ transfer, zero overlap ⇒ no transfer.

7 Discussion

Excitatory vs. inhibitory gender features. The most important finding is the direction divergence across architectures. PaLiGemma-3B (a simpler, captioning-focused model) appears to encode gender through *excitatory* features: these features directly contribute gendered language to the output, so ablation reduces it. Qwen2-VL-7B and Llama-3.2-11B (larger, instruction-tuned) appear to use *inhibitory/regulatory* features: these features modulate or suppress gendered output, so ablation releases it.

This finding has practical implications: a naïve debiasing approach that ablates top-differential features would *reduce* gender bias in PaLiGemma-3B but *increase* it in Qwen2-VL-7B/Llama-3.2-11B. Architecture-specific analysis is essential before deploying SAE-based debiasing interventions.

Architecture hypothesis. The direction divergence—excitatory in PaLiGemma-3B, inhibitory in Qwen2-VL-7B/Llama-3.2-11B—may reflect fundamental differences in how models learn to handle gendered language. We hypothesize that instruction-tuned models develop regulatory circuits that suppress gender expression during alignment, while captioning-focused models directly produce gendered descriptions. Both instruction-tuned models use identical hooks and show identical direction, supporting an architectural rather than methodological explanation. We use vanilla ReLU SAEs as they remain the most widely deployed architecture and serve as a conservative baseline.

Hook method confound. We acknowledge that the direction difference could partly reflect the different hook methods (full reconstruction vs. residual ablation). However, both Qwen2-VL-7B and Llama-3.2-11B use the same residual hook and show the same direction, while the full-reconstruction hook *destroyed* Qwen2-VL-7B's output entirely—suggesting the direction difference is primarily architectural. Running PaLiGemma-3B with the residual hook would disambiguate this (see Limitations).

Cross-lingual implications. The near-zero feature overlap across all models challenges the assumption that multilingual models share bias representations across languages. Our causal cross-lingual ablation (Table 9) provides direct evidence: for Qwen2-VL-7B (0% feature overlap), ablating English gender features has *no effect* on Arabic captions and vice versa, demonstrating that debiasing in one language does not transfer. Critically, PaLiGemma-3B (57% feature overlap) shows the opposite—Arabic features *do* transfer to English—confirming that the degree of cross-lingual transfer is directly predicted by feature overlap. This implies that multilingual fairness requires auditing feature overlap before assuming transfer, and likely necessitates language-specific debiasing interventions for models like Qwen2-VL-7B and Llama-3.2-11B.

Mean-pooling trade-off. Our SAEs are trained on mean-pooled activations but applied to per-token activations during generation. For PaLiGemma-3B, this works because the SAE has near-perfect reconstruction on pooled data. For Qwen2-VL-7B/Llama-3.2-11B, the mismatch necessitated residual ablation (Eq. 8). Training per-token SAEs would avoid this mismatch but at significantly higher computational cost.

8 Limitations

1. **SAE training scale:** SAEs trained on 5K–10K samples per model; larger training sets may yield better features.
2. **SAE architecture:** We use vanilla ReLU SAEs; TopK or JumpReLU architectures may improve feature quality.
3. **Evaluation:** We do not report SAEBench (Karvonen et al., 2025) standardized metrics; future work should use this benchmark.
4. **Binary gender:** We track binary gender terms only; non-binary terms are tracked separately but not used as the primary metric.
5. **Hook inconsistency:** PaLiGemma-3B uses full reconstruction; Qwen2-VL-7B/Llama-3.2-11B use residual ablation. Running PaLiGemma-3B with residual ablation would test whether the direction difference is architectural or methodological.
6. **No human evaluation:** We measure gender term counts but do not evaluate caption fluency or semantic accuracy after ablation. Human evaluation of caption quality is orthogonal to our research question of causal feature identification.
7. **Feature polysemanticity:** We cannot claim the ablated features *exclusively* encode gender; they may be polysemantic.
8. **Llama SAE quality:** Llama-3.2-11B's SAE has only 36.6% explained variance; however, the finding that intervention still produces significant effects suggests our approach may be conservative—better-quality SAEs would likely yield larger effects.

9 Conclusion

We demonstrate that sparse autoencoders can identify and causally intervene on gender bias features in VLM representations. Three key findings emerge: (1) targeted ablation of just 0.6% of features produces 1.8–7.1× larger effects than random ablation across all models tested; (2) different architectures encode gender through fundamentally different mechanisms—excitatory in captioning-focused models vs. inhibitory in instruction-tuned models; (3) cross-lingual feature overlap directly predicts whether debiasing interventions transfer across languages. These findings have immediate practical implications: architecture-specific and language-specific analysis is essential before deploying SAE-based bias mitigation.

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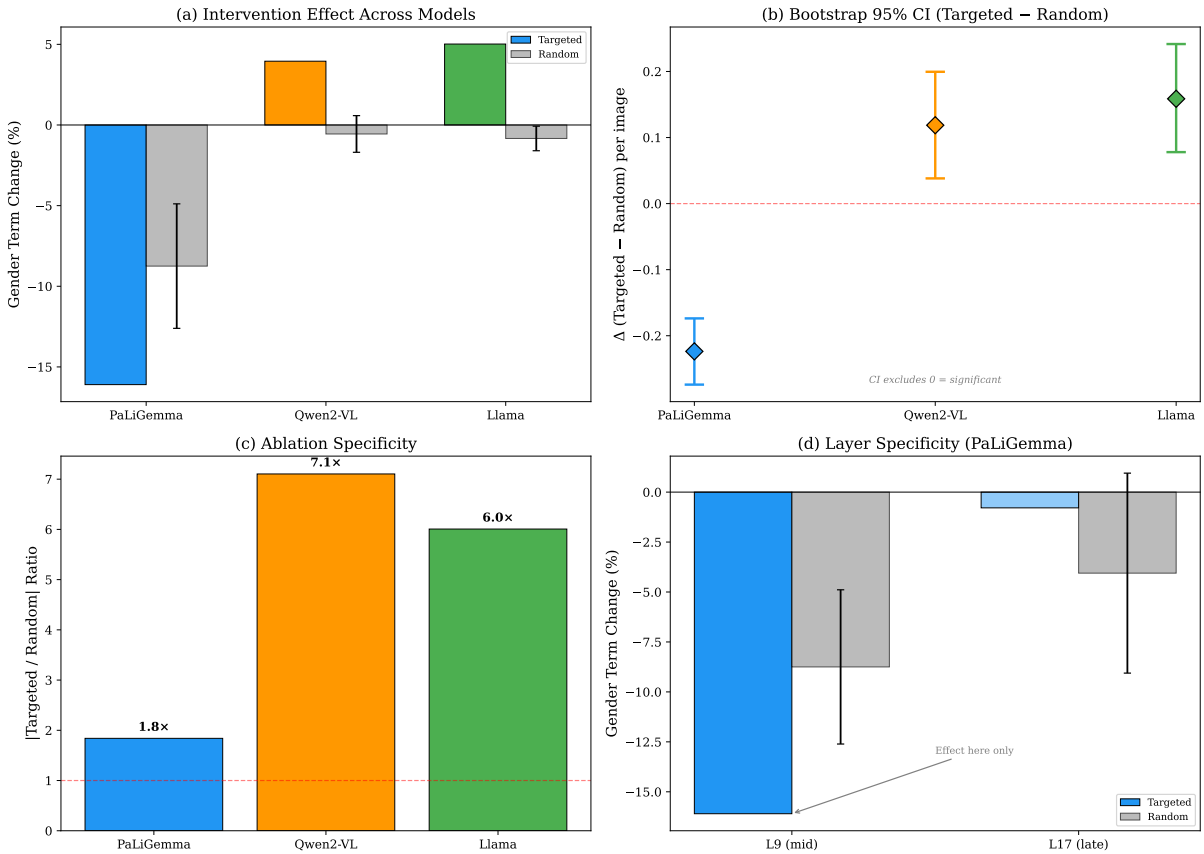

Figure 8: Comprehensive summary of main findings. (a) Cross-model intervention: targeted ablation of 100 gender features produces significant changes in all three models, with opposite directions for excitatory (PaLiGemma-3B) vs. inhibitory (Qwen2-VL-7B, Llama-3.2-11B) architectures. (b) Bootstrap 95% CIs confirm significance. (c) Effect specificity ratios. (d) Layer specificity in PaLiGemma-3B: gender features are concentrated at L9.

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A Additional Per-Term Analysis

Complete per-term breakdown for all models is available in the supplementary materials and at `https://github.com/nour-mubarak/mechanistic_intrep`.

B SAE Training Details

Table 10: SAE training hyperparameters.

Parameter	**Value**
Expansion factor	8×
L1 coefficient (λ)	10^{-4} (PaLiGemma-3B), 5×10^{-4} (Qwen2-VL-7B)
Learning rate	10^{-4}
Optimizer	AdamW
Batch size	256
Epochs	50 (early stopping, patience=10)
Warmup steps	1,000
Train/Val split	90/10

C Random Ablation Distributions

D SAE Quality vs. Effect Size

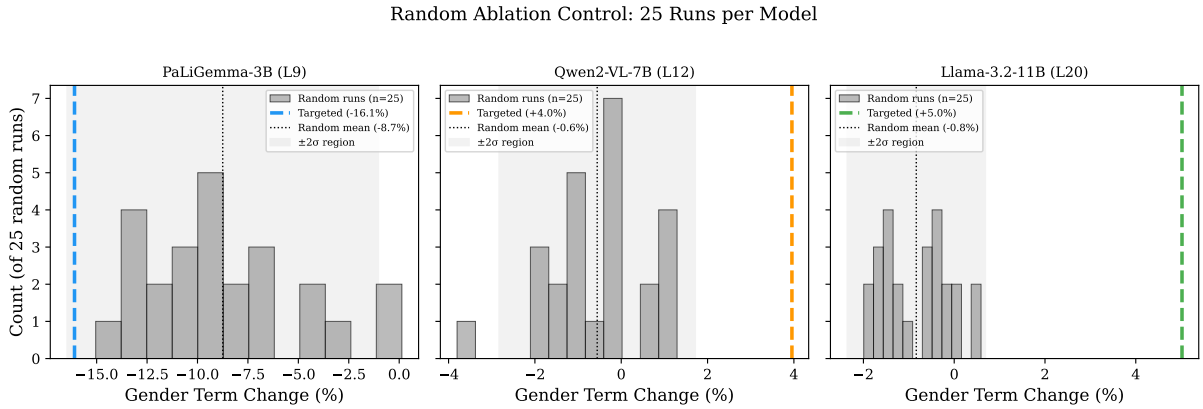

Figure 9: Distribution of gender term changes across all 25 random ablation control runs for each model. The targeted ablation effect (red dashed line) falls well outside the random distribution in all three cases, confirming specificity.

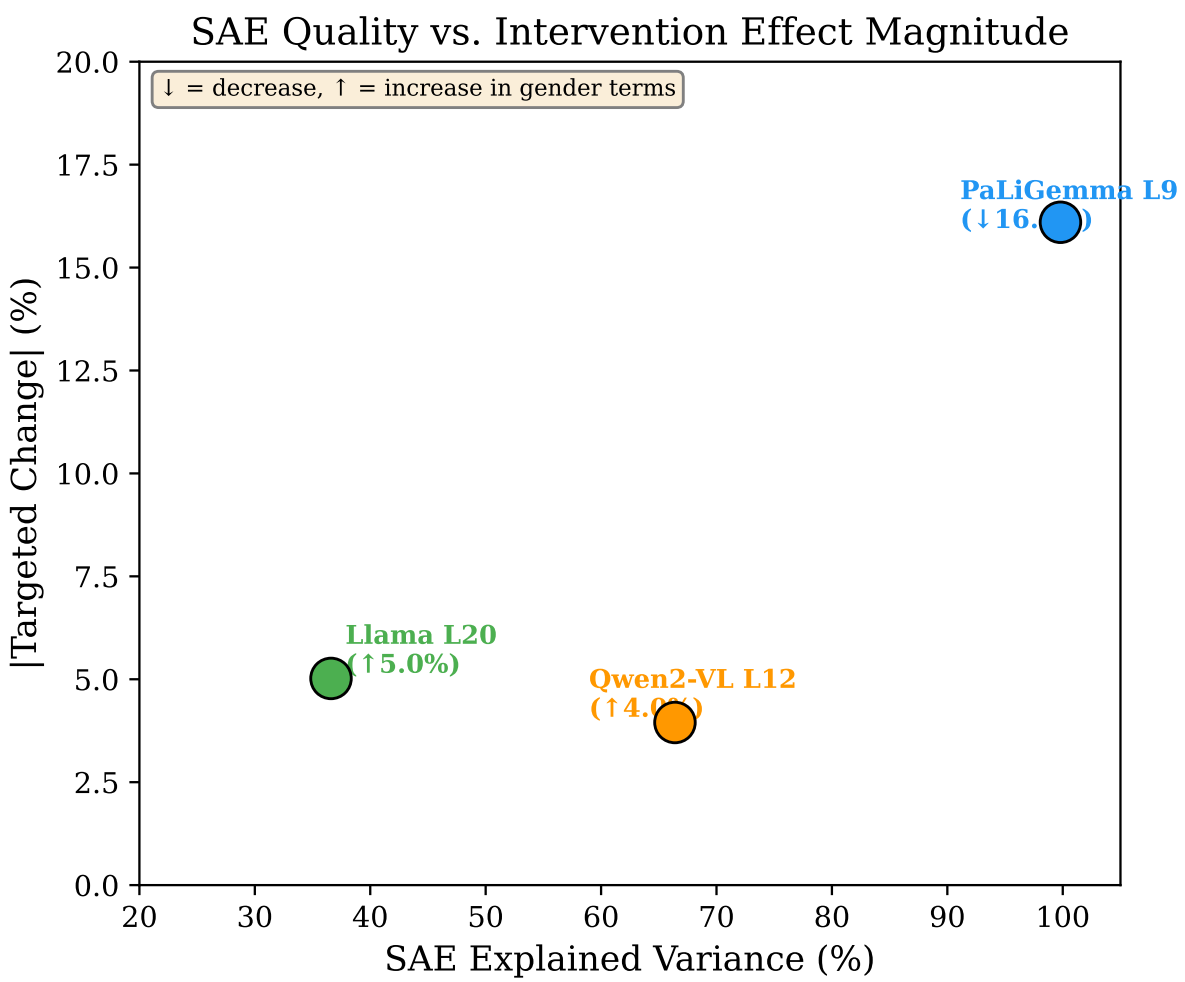

Figure 10: SAE quality (explained variance) vs. intervention effect magnitude. Despite large differences in SAE quality across models, all produce significant intervention effects—suggesting the approach is robust to SAE training quality.